1.1 B ✓✓
1.2 C ✓✓
1.3 D ✓✓
1.4 C ✓✓
1.5 C ✓✓
1.6 B ✓✓ **[2x6=12]**

2.1.1 F ✓ (1)
2.1.2 D ✓ (1)
2.1.3 A ✓ (1)
2.1.4 C ✓ (1)
2.1.5 E ✓ (1)
2.1.6 NH_4NO_3 ✓✓ (2)

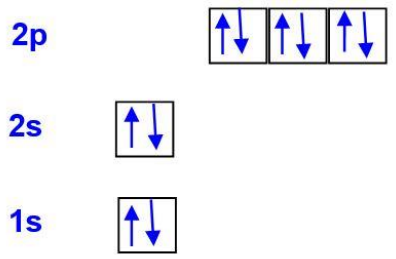
2.2.1 Calcium✓ (1)
2.2.2 Noble gases ✓ (1)
2.2.3 P ✓ (1)
2.2.4 Ca✓$C\ell_2$✓ (2)

2.3.1 neutrons ✓ (1)

2.3.2 Similar: Both have 5 protons ✓
Difference: B-10 has 5 neutrons and B-11 has 6 neutrons ✓ (2)

2.3.3 (19% x 10) + (81% x 11) ✓m = 10,81✓ amu✓ (3)

2.3.4



✓ 10 electrons
✓ correct orientation of arrows
✓ labelled energy levels (3)

[21]

3.1.1 ionic ✓ (1)

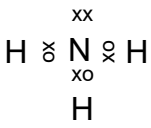
3.1.2 polar ✓ covalent ✓ (2)

3.2 Unequal ✓ sharing ✓ of electrons ✓ (3)

3.3 Fe ✓ (1)

3.4 ammonia ✓ (1)

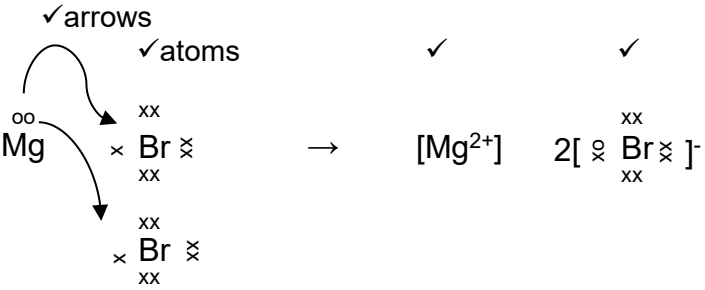
3.5



✓✓ (2)

3.6 Hydrogen ✓ bonds ✓ (2)

3.7



(4)
[16]

4.1.1 Decomposition ✓ (1)

4.1.2 A glowing ✓ splint will reignite. ✓ (2)

4.1.3



✓✓ (2)

4.1.4 London ✓ dispersion forces.
A temporary ✓ dipole forms in a molecule which induces ✓ a dipole in the adjacent molecule. (3)

4.1.5 $n_{H2O2} = \frac{m}{M} = \frac{20,4}{34\checkmark} = 0,6\ mol\checkmark$

$mol\ ratio\ \ H_2O_2 : H_2O$
$2 : 2 \quad \checkmark$

$m_{H2O} = nM = 0,6(18)\checkmark = 10,8\ g \quad \checkmark$ (5)

4.1.6 $mole\ ratio\ \ H_2O_2 : O_2$
$2 : 1$
$0,6 : x$
$n_{O2} = \frac{0,6}{2} = 0,3\ mol \quad \checkmark$

$V_{O2} = nV_m = 0,3(22,4)\checkmark = 6,72\ dm^3\checkmark$ (3)

4.2.1 2✓K + $CuSO_4 \rightarrow Cu + K_2SO_4$ (1)

4.2.2 Copper (II) ✓ sulfate ✓ (2)

4.2.3 Displacement ✓ (1)

4.2.4 K_2SO_4: purple/lilac ✓
$CuSO_4$: blue/green ✓ (2)

4.2.5 $CuSO_4(s) \rightarrow Cu^{2+}(aq)$ ✓ + $SO_4^{2-}(aq)$✓ ✓state symbols (3)

4.3.1 1. $BaC\ell_2$ ✓ white precipitate forms✓
2. $HC\ell$ ✓ precipitate remains✓

Or 1. $HC\ell$ no bubbles
2. $BaC\ell_2$ white precipitate forms (4)

4.3.2.1 lead (II) ✓ nitrate✓ (2)

4.3.2.2 canary (bright) yellow precipitate ✓ (1)

4.3.2.3 $Pb(NO_3)_2(aq) + 2LiI$✓ $(aq) \rightarrow PbI_2$✓ $(s) + 2LiNO_3$✓ (aq) ✓bal ✓sym (5)
[37]

5.1.1 sodium✓ hydrogen carbonate✓ / sodium bicarbonate (2)

5.1.2 clear✓ lime water✓ turns milky✓ (3)

5.2.1 +✓ 5✓ (2)

5.2.2 +✓ 7✓ (2)

5.3.1 Loss of electrons. ✓✓ (2)

5.3.2 i) H_2 ✓✓ (2)

ii) Br_2 ✓✓ (2)

iii) H^+ ✓✓ (2)

5.3.3 Oxidation number of H in H_2 is 0. ✓
Oxidation number of H in $HC\ell$ is +1. ✓
Therefore loss of electrons ✓has occurred. (3)
[20]

6.1 $Na_2CO_3 + \mathbf{2}$✓$HC\ell \rightarrow \mathbf{2}$✓$NaC\ell + H_2O + CO_2$ (2)

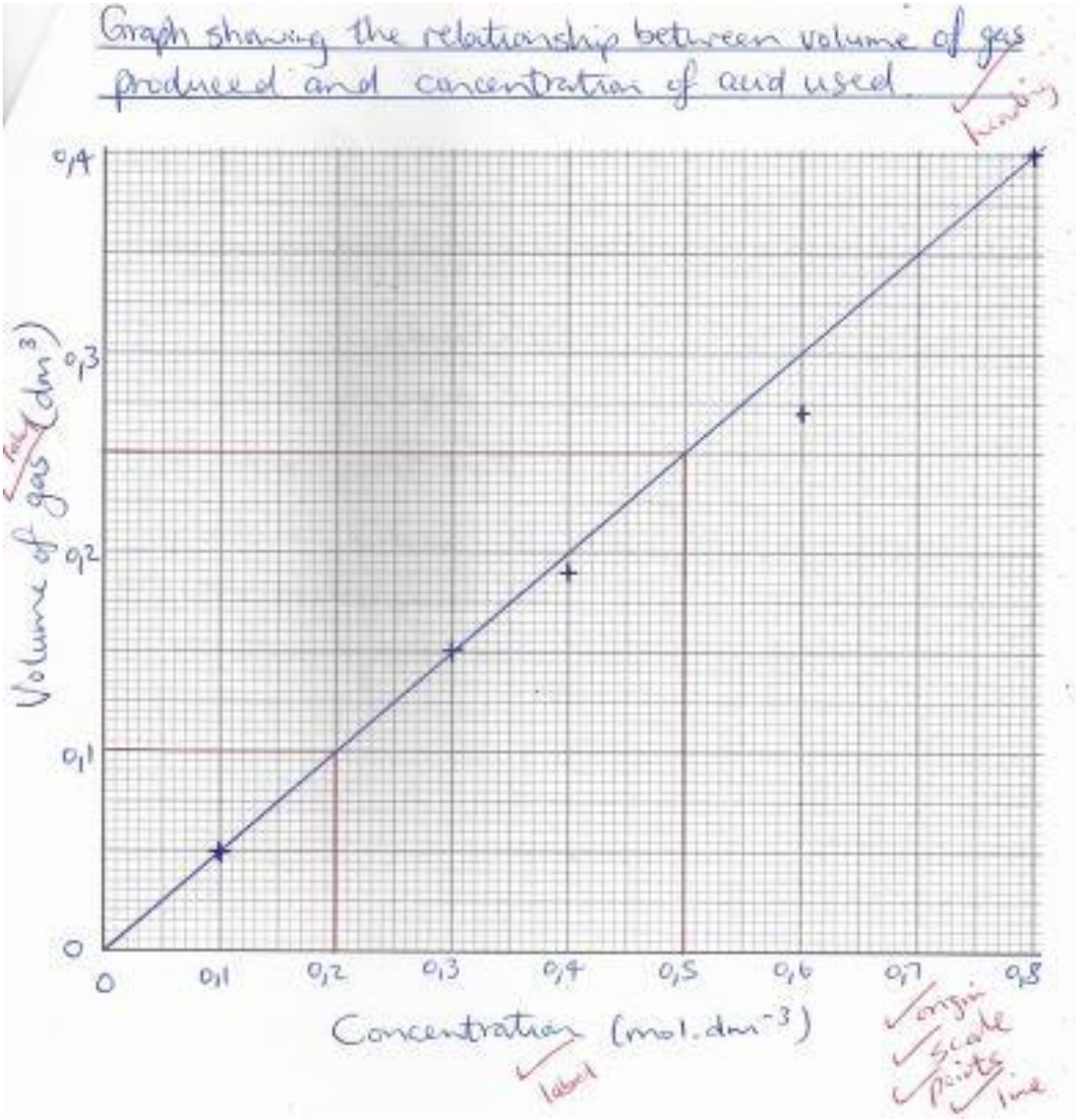
6.2 $\frac{0,26+x+0,29}{3} = 0,27$
$x = 0,26$ ✓ (1)

6.3 Volume of gas. ✓ (1)

6.4 Same mass of Na_2CO_3 ✓ Must say SAME/FIXED
Same temperature ✓
Same volume of acid
Same time period (2)

6.5 In order for **a fair** ✓test, only one variable✓ may be changed. Could describe how the two above would have change to outcome if not fixed. (2)

6.6



(7)

6.7 Volume✓ of gas produced is directly proportional✓ to the concentration✓ of the acid. (3)

6.8 $m = \frac{0,25-0,1✓}{0,5-0,2✓} = 0,5✓$ Max one out of 3 if used plotted values. Must read off values from line of best fit. (3)

6.9 Water contains hydrogen bonds. ✓
Carbon dioxide has London dispersion forces. ✓
Hydrogen bonds are stronger ✓than London dispersion forces.
Therefore more energy ✓would be needed to change water to a gas, room temperature does not have sufficient energy to change water to a gas. (4)

[25]